Comments in Response to:

Development of an Artificial Intelligence (AI) Action Plan

Dan Stanzione, Associate VP for Research, The University of Texas at Austin

Since 2021, the investment in AI infrastructure from the private sector is unrivaled by any other technological investment in history. In the current year alone, investment from the five largest companies in datacenters and servers for AI totals more than $300 Billion USD, with total worldwide investment substantially higher. Cumulative investment is well over a trillion dollars and rising quickly, and this does not include the massive investments in human capital associated with this infrastructure.

In this environment, it seems difficult for a federal investment to "move the needle" on AI for the largest corporations, beyond insuring the supply chain from impacts by other state actors, and providing sufficient safeguards for the public in the use of AI. However, there are many gaps in the current environment that are crucial to US economic development and US national security that could be addressed by a federal investment in AI infrastructure.

With state-of-the-art models for Generative AI limited to only a few of the very largest players, the following gaps clearly exist:

- The need to develop a broad pipeline to provide the future AI workforce, and AI-enabled workforce, to supply the needs of the American economy.
- The need to democratize access to AI infrastructure, not only to develop the workforce but to spur innovation and research across small and medium enterprises, across academia, and broadly across government.
- The need to develop trusted and open models that can be deployed effectively by the public, and by the public sector, to make sure the benefits of AI are available for a broad cross section of societal needs, and not simply for the most profitable ones.

To meet these needs, a three pronged government approach investing in AI infrastructure, AI education and training, and AI research and innovation is needed.

The infrastructure needs and a plan of action is well-documented, with many excellent ideas codified in the reports on NAIRR (the National AI Research Resource) and in the proposed Department of Energy FAST initiative. The government should invest in a shared national cyberinfrastructure: a broadly accessible and federated collection of resources critical for AI research, including computational resources, public- and private-sector data, software, and testbeds. This resource would be intended to serve U.S.-based AI researchers, students, and entrepreneurs who are pursuing foundational, use-inspired, and translational AI research. The components of this infrastructure would include:

- Compute: provide access to the diversity of computing resources that fuel AI R&D. The infrastructure would ideally consist of a federated mix of on-premise and commercial computational resources, including conventional servers, computing clusters, high-performance computing, and cloud computing, and also support access to edge computing resources.

- Data sets: The infrastructure would either fund the creation of and/or provide continuing support to existing AI data repositories. Data from these providers would be easily found and accessed through a central catalog and be "ready to ingest" by virtue of interoperability standards for such data repositories set by the infrastructure providers. The infrastructure should include a “data marketplace” where data resources could be contributed by users or stakeholders. Contributions would be subject to dataset acceptance criteria for quality.

- Testbeds: The infrastructure would act as a hub for AI testbeds and test datasets, cataloging and providing access to existing testing and benchmarking resources that can accelerate progress in AI capabilities and increase research community access to high-quality testbeds.

- User training and services: Software, training, and educational resources would be made available to support a diverse set of users with varying levels of proficiency. The infrastructure would provide multiple levels of user support including help desk and solution consulting, and cultivate communities of practice to provide peer-to-peer support.

To maximize the use of this infrastructure, substantial corresponding investments should also be made in education and research programs. The education program should span the University, community college, and K-12 levels to both develop curriculum and prepare a workforce of both creators and consumers of AI tools and models.

The research program should be similarly broad. A program is needed that supports fundamental research in AI – development of new tools, models, hardware, and approaches for the foundations of AI, for ways to make AI transparent and trustworthy, and ways to insure security and privacy with AI. There should also be support for use-inspired research in translational AI, to apply the best tools and practices of AI across novel applications, both for commercial purposes, and in areas of interest to society and the public sector, such as creating more efficient government, application to science and knowledge advancement, national security, etc.

The potential impacts of AI – economically, scientifically, and societally – are well-documented. The US private sector has made enormous investments in the economic opportunities presented. The geopolitical rivals of the US have made clear, focused investments in AI, in the technologies supporting AI (e.g. semiconductors, computer architectures, etc.), and in the applications of AI. It is critical that the US government represent not only the opportunities in AI, but the risks and costs of being outcompeted by rival states to develop and use this technology. A well-balanced public investment that complements the work of the private sector in this area can help ensure a prosperous and secure future for the United States.